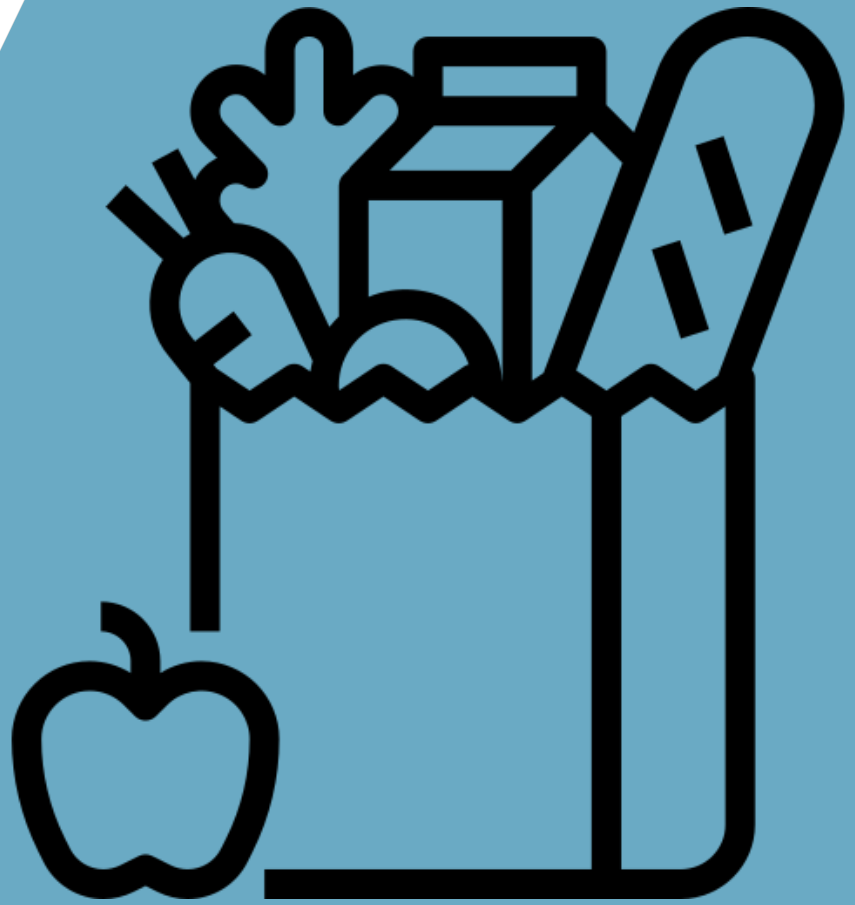
MARKETING ANALYTICS PROJECT WORK

CUSTOMER RELATIONSHIP MANAGEMENT



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Introduction and Objectives

CLIENT	COOP
OBJECTIVES	Understanding the **purchase behaviour of customers** Understanding the **loyalty mechanisms retailer-customer**
OUTPUT	Tailored **marketing Strategies** and **recommendations**

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DATA INSPECTION | TYPE A

1 General understanding of the dataset:

VARIABLE	TYPE	VARIABLE	TYPE
Item_id	Numerical	customer_id	Categorical
price per unit	Numerical	brand_position	Categorical
price per kg	Numerical	sector	Categorical
DATE	Numerical	family	Categorical
UNITS	Numerical	category	Categorical
KG	Numerical	brand	Categorical
		sub-brand	Categorical

2 Construction of the **Anomaly Table** (see table below) to highlights possible data problems.

3 Focus on the most relevant problem, **the missing value in the quantity purchased**. Relative rows were **deleted**.

ANOMALY	COUNT	PERCENTAGE
NEGATIVE VALUES IN UNITS	0	0
NEGATIVE VALUES IN KG	0	0
NEGATIVE VALUES IN PRICE.PER.UNIT	0	0
NEGATIVE VALUES IN PRICE.PER.KG	0	0
NA VALUES IN SUB.BRAND	192357	17.59141303
NA VALUES IN BRAND	106466	9.736517932
NA VALUES IN CATEGORY	0	0
NA VALUES IN FAMILY	0	0
NA VALUES IN SECTOR	0	0
NA VALUES IN BRAND_POSITION	1535	0.140378666
BOTH UNITS AND KG ARE EQUAL TO 0	10	0.000914519

4 Missing values in the sub.brand and brand were filled by analysing similar products types and acquiring the missing data information.

MARKET BASKET ANALYSIS | 1

GOAL

To analyse consumption dynamics and determine whether different customer segments exhibit unique shopping behaviours, distinct MBAs were conducted. For this purpose, RFM analysis was first performed on dataset type A to create meaningful clusters.

RFM scores were used as input for clustering algorithms. Then, the optimal cluster count (6) was determined using silhouette scores and the elbow method. Finally, clusters were labeled based on behavioral patterns observed in the RFM metrics. The six clusters are: Loyal Low-Value Customers, Engaged High-Value Customers, Potential High Spenders, Frequent Moderate Buyers, Occasional High Spenders, Irregular Low-Value Customers

OBSERVATION

The results from individual cluster analyses were similar across all segments.

⇒ The assumption that clusters would exhibit distinct consumption patterns did not hold strongly in this case.

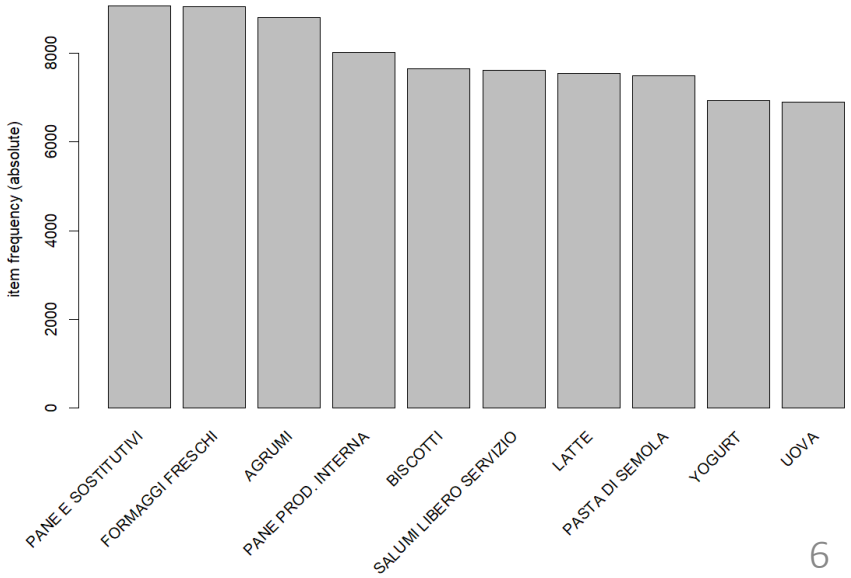
DECISION

To streamline the process and derive actionable insights efficiently, we proceeded with **MBA on the complete dataset type A**.

1. The dataset was grouped by `Customer_id` to create a list of transactions:
   ⇒ Each transaction consists of the `CATEGORY` column values for a given customer.
   This list format was converted into a `transactions` object, suitable for association rule mining using the `arules` package.

2. The relative frequency of each item in the dataset was calculated using the `itemFrequency()` function.

3. Items were **sorted by frequency** to identify the most purchased products. This provided insights into which items were the most frequent among customers.



MARKET BASKET ANALYSIS | 2

4 The **Apriori algorithm** was applied to the transactions dataset with the following parameters:

- **Support**: minimum threshold of 0.01.
- **Confidence**: minimum threshold of 0.65.
- **Maximum Rule Length**: restricted to 4 items.

The generated rules were sorted by `lift` (a measure of association strength), and the top rules were inspected (See *Annex 3*).

5 To focus on meaningful patterns, rules were filtered based on minimum thresholds for `lift`, `support`, and `confidence`.

- **Lift > 1.2**. Higher and positive Lift shows that the antecedent itemset can predict the presence of the consequent itemset.
- **Support > 0.02**
- **0.65 < Confidence ≤ 0.8**. It is important that Confidence is high enough to have a meaningful rule but not too high otherwise the potential development deriving from this rule decreases.

6 For each unique `rhs` (right-hand side), the top 5 rules were selected based on `support` and `lift` and `confidence`. The filtered rules were stored in a consolidated list of "top rules" (See *Annex 1*).

7 For each `rhs`, the associated `lhs` items were summarized into a list to understand which items lead to the purchase of specific products. The most frequently occurring `lhs` items for each `rhs` were identified and ranked.

RULES	LHS	RHS
{BANANE} => {YOGURT}	{BANANE}	{YOGURT}
{BIRRA} => {SNACK SALATI}	{BIRRA}	{SNACK SALATI}
{CEREALI PRIMA COLAZIONE} => {YOGURT}	{CEREALI PRIMA COLAZIONE}	{YOGURT}
{CONDIMENTI} => {PASTA FRESCA}	{CONDIMENTI}	{PASTA FRESCA}
{PREPARATI PER DOLCI} => {FARINA, LATTE}	{PREPARATI PER DOLCI}	{FARINA, LATTE}
{PRODOTTI BAGNO} => {IGIENE ORALE}	{PRODOTTI BAGNO}	{IGIENE ORALE}
{SALUMI A SERVIZIO ASSISTITO} => {PANE PROD. INTERNA}	{SALUMI A SERVIZIO ASSISTITO}	{PANE PROD. INTERNA}
{TRATTAMENTO CORPO/VISO} => {IGIENE ORALE}	{TRATTAMENTO CORPO/VISO}	{IGIENE ORALE}
{SURGELATI BASE FARINA} => {BIBITE, SURGELATI VEGETALI}	{SURGELATI BASE FARINA}	{BIBITE, SURGELATI VEGETALI}
{BISCOTTI} => {LATTE}	{BISCOTTI}	{LATTE}
{CONSERVE POMODORO} => {PASTA DI SEMOLA}	{CONSERVE POMODORO}	{PASTA DI SEMOLA}
{SURGELATI PESCE} => {SURGELATI VEGETALI}	{SURGELATI PESCE}	{SURGELATI VEGETALI}
{UOVA} => {ALTRI ORTAGGI}	{UOVA}	{ALTRI ORTAGGI}

* For a complete visualisation of all the rules and the metrics, please see to *Annex 2*.

DATA INSPECTION | TYPE B

1 General understanding of the dataset:

2 Construction of the **Anomaly Table** (see table below) to highlight possible inconsistencies in the dataset

3 Here we can see that specific transactions have a gross PL sale higher than the net sales. In our opinion, the PL_gross_sales is higher than the net sales because it is not discounted. This will however not affect our churn output because from a monetary point of view we will use gross_sales for our calculations.

VARIABLE NAME	VARIABLE TYPE
ticket_id	Numerical
id_customer	Numerical
store_id	Numerical
store_type	Categorical
PL_gross_sales	Numerical
net_sales	Numerical
gross_sales	Numerical
number_items_other	Numerical
number_item_PL	Numerical

ANOMALY	COUNT	PERCENTAGE
Negative values in net_sales	0	0
Negative values in gross_sales	0	0
Negative values in PL_gross_sales	0	0
If number_item_PL is NA, PL_gross_sales different from 0	0	0
PL_gross_sales > net_sales	189285	9.139051
PL_gross_sales > gross_sales	0	0
net_sales > gross_sales	2	9.66E-05
Both number_item_PL and number_items_other are equal to 0	0	0
NA values in net_sales	0	0
NA values in gross_sales	0	0
NA values in PL_gross_sales	353544	17.0698
NA values in number_item_other	47813	2.308505
NA values in number_item_PL	353544	17.0698

RFM ANALYSIS

OBJECTIVE

Identifying and excluding outliers

Remove customers with extreme values in **Recency**, **Frequency**, and **Monetary Value** to prevent skewed clustering.

PROCESS

IQR Method

- Calculated the **Interquartile Range** (IQR) for each metric.
- Defined upper bounds as Q3+k*IQRQ3+k*IQR, with k=2k=2 determined as the optimal multiplier (explained below).

Dynamic Percentile

- Explored percentiles (70th to 99th) to dynamically refine thresholds.
- Used the **Elbow Method** to identify where the increase in outliers flattens, ensuring meaningful thresholds.

Multiplier Analysis

- Tested kk values from 1 to 3, incrementing by 0.5.
- Plotted the number of outliers against kk, selecting k=2k=2 as the point where outliers stabilized.

IMPACT

This process ensured the removal of extreme cases, such as infrequent but high-value purchases or highly recent yet overall low-engagement customers, creating cleaner and more actionable data.

Key **RFM metrics** were calculated for each customer:

- **Recency**: Days since the last purchase.
- **Frequency**: Number of distinct purchase dates.
- **Monetary Value**: Total gross sales per customer.

Additional features included:

- **Average Inter-Purchase Time (AVG_IPT)**: Average time between purchases.
- **Average and Standard Deviation of Ticket Value**: Metrics indicating spending consistency.

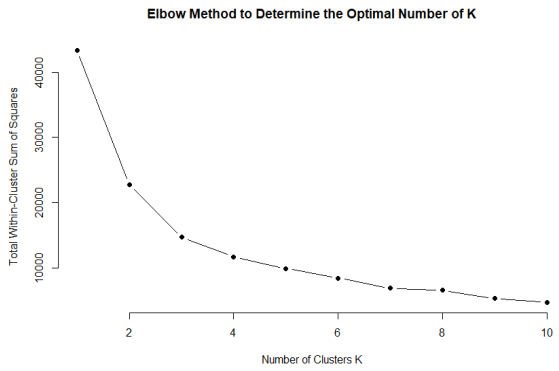
Reducing the Number of Clusters

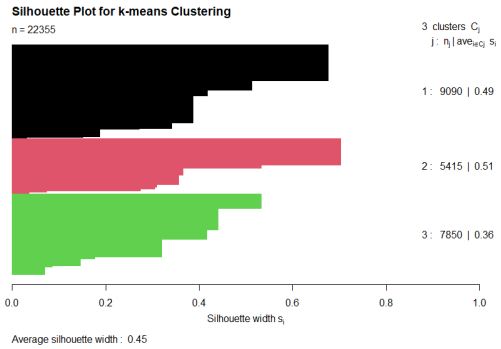
a. PCA (Principal Component Analysis)

PCA was used to transform the three RFM variables (Recency, Frequency, and Monetary) into a smaller set of uncorrelated components:

- **PC1** captured overall engagement, with contributions from all three variables.
- **PC2** highlighted dynamics like Recency versus spending.

This allowed us to visualize and cluster the data in a 2D space while retaining most of the original information.



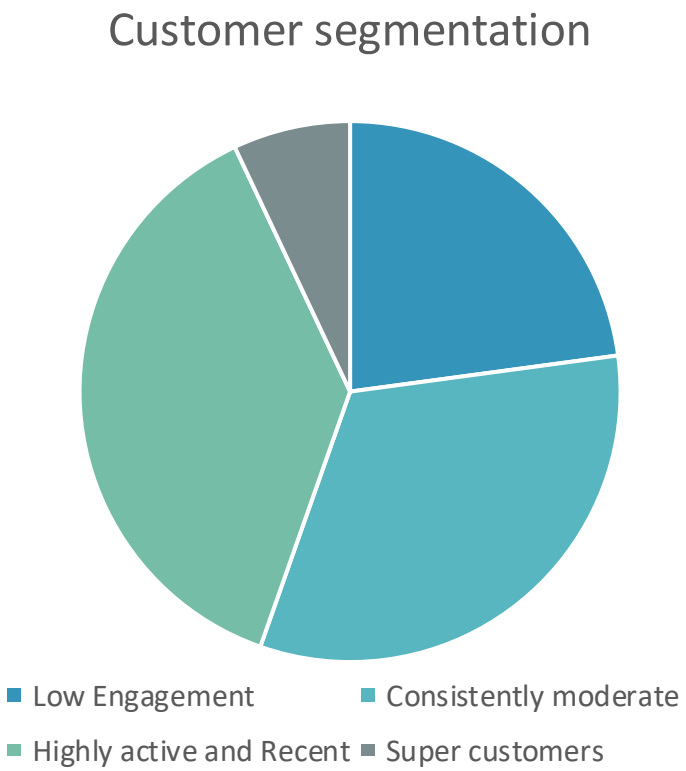


b. K-Means Clustering

To identify natural groupings among customers, we applied **K-Means clustering** on the RFM scores. The process involved:

1. Running the **Elbow Method** to determine the optimal number of clusters by examining the within-cluster sum of squares (WSS).
2. Testing different values of **K** and analyzing clustering quality using **silhouette scores**, which measure how well-separated the clusters are.

- Customers were labeled based on their RFM scores:
  - Recency, Frequency, and Monetary Value were categorized as Low, Medium, or High using quartiles.



CUSTOMER SEGMENTATION | 2

Consistently Moderate

Behavior:
Moderate average ticket value and purchase frequency.
Recent purchases but not the most actively engaged.
Insight:
Represents a stable revenue stream.
Opportunity to increase loyalty through tailored offers

High and Recently Active

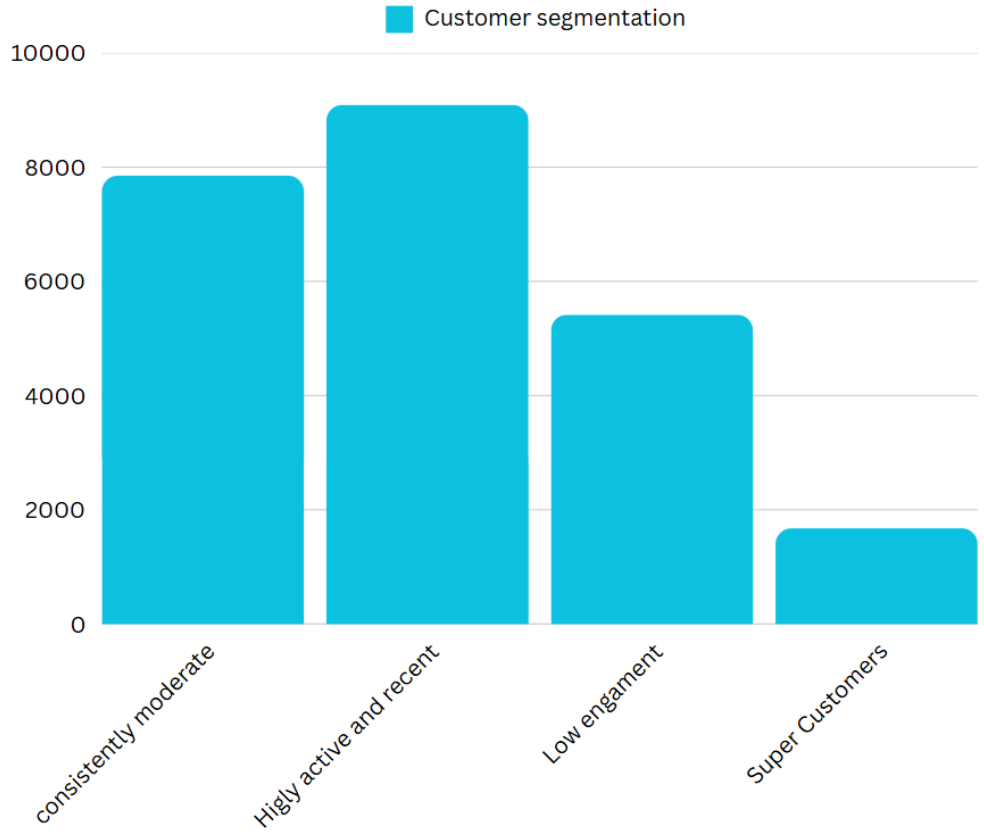
Behavior:
Moderate ticket value with high purchase frequency.
Very recent purchasers, highly engaged.
Insight:
Valuable customers due to active engagement.
Retention strategies should be prioritized.

Low Engagement

Behavior:
Low average ticket value and purchase frequency.
Less recent purchases indicating inactivity.
Insight:
Potential reactivation through personalized promotions.

Super Customers

Behavior:
Highest average ticket value and purchase frequency.
Regular and recent purchasers.
Insight:
Most valuable customers.
Should receive exclusive offers to maintain loyalty.



Additional Observations Private Label (PL) Product Purchases: PL product purchase percentage is consistent across clusters (28%–32%). Indicates uniform interest in PL products regardless of customer segment.

CHURN ANALYSIS | Churn Definition

STARTING CONCEPT

The analysis focuses on **benchmarking** each **customer's behavior** dynamically by comparing their **current performance** to their own **historical patterns**. Using **confidence intervals**, avoids rigid thresholds, accounting for variability in individual spending habits. This ensures **fair evaluation** even for customers with irregular purchase patterns

TIME BASE BEHAVIOR

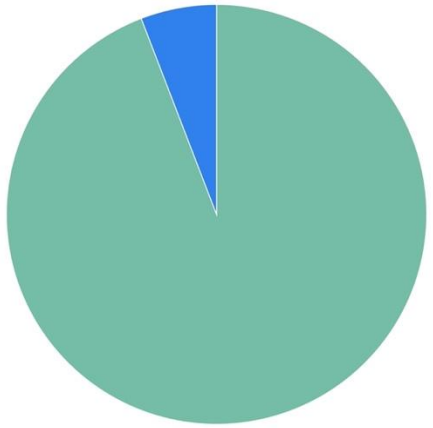
The dataset is segmented into **31-day** slots to capture monthly behavioural trends for key metrics. This temporal segmentation enables the identification of **evolving trends** while smoothing out short-term fluctuations. **Rolling and cumulative metrics** ensure a balance between immediate trends and long-term patterns, preventing premature churn detection due to temporary anomalies.

PROACTIVE CHURN DETECTION

The system is designed to detect churn early by identifying **sustained underperformance over the 3 past periods** or **extended inactivity**. Customers are flagged only when clear signs of disengagement emerge, ensuring **retention efforts are proactive rather than reactive**. Special care is taken to include customers with limited transaction data or sporadic activity, applying approximations where necessary to avoid misclassification.

ALIGNMENT WITH BUSINESS PRIORITIES

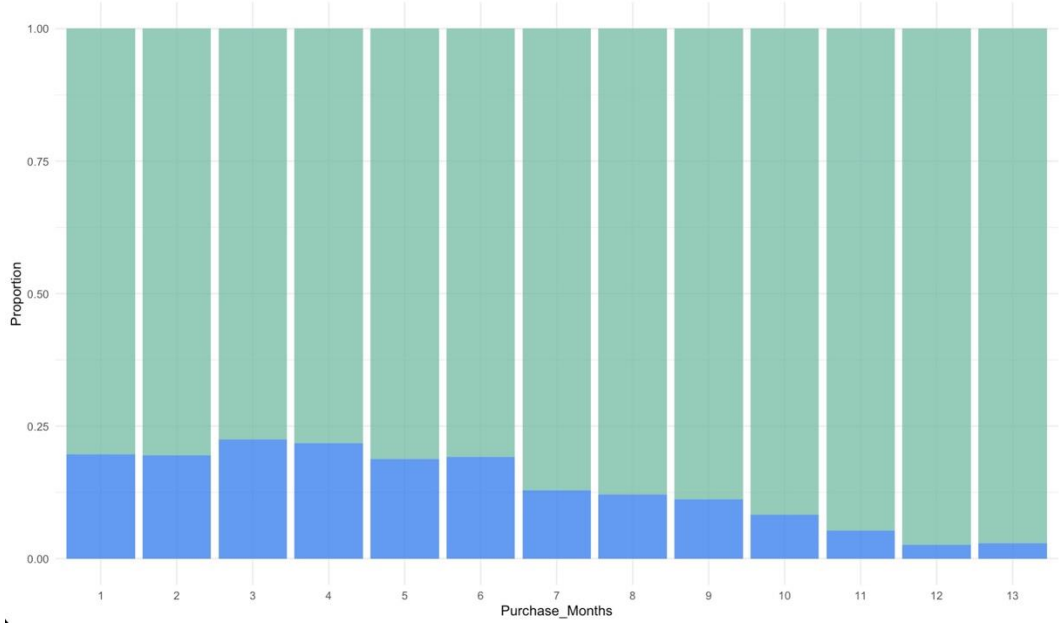
The approach emphasizes **revenue-centric analysis**, prioritizing customers whose declining behaviour poses a financial risk to the business. By focusing on different variables, the analysis **aligns churn detection with operational and strategic goals**, allowing businesses to target high-value customers with tailored retention strategies for maximum impact.



Churned

Not Churned

Proportion of Churned vs Not Churned



Churned

Not Churned

Churn Customers proportion on Purchase_Months

CHURN ANALYSIS | Churn Modelling

Three models for churn defection were chosen to provide a comprehensive evaluation of the factors influencing churn.

1

LOGISTIC REGRESSION

Offers interpretability and highlights linear relationships.

RANDOM FORREST

Captures complex non-linear interactions and ranks variable importance through ensemble learning.

XGBOOST

Provides state-of-the-art performance with advanced boosting techniques to handle intricate patterns.

2

VARIABLE AGGREGATION

We combined the variable importance output of the three models to identify the ones upon which to focus our attention.

	Rank	Logistic Regression	Random Forrest	XGBoost	Impact on Churn
Final_Segment	1	1	1	1	Depends of the Cluster
Purchase_Months	2	6	2	2	-0.2199
PL_ratio	3	2	4	10	1.8851
Frequency	4	3	5	9	-0.5601
STD_Ticket	5	7	6	5	-0.2404
Montetary_Value	6	4	9	7	-0.3336
AVG_item_PL	7	5	3	14	-0.2648

3

VARIABLE INTERPRETATION

Frequent buyers demonstrate **higher engagement** with the brand, which fosters loyalty.

Loyalty builds over time, with long-term customers being more stable and reliable.

Customers heavily reliant on private-label products may be more price-sensitive and vulnerable to competitor offerings.

Variability suggests a diverse shopping pattern, indicating a broader relationship with the brand.

STATISTICAL INSIGHTS | Private Label ANALYSIS ON LOYALTY

The relevance and the impact of **PL_ratio** and **AVG_item_PL** is combined to offer a broader perspective of the impact of Privat Label purchase on the loyalty of the customers.

1 OUTPUT AGGREGATION

I aggregated the output of the three models, to have a better understanding of the impact:

- Output of the **Logistic Regression**
- **Partial Dependence Plot** for **Random Forrest** and **XGBoost**

2 OUTPUT INTERPRETATION

The output brings us to the definition of **two specific cases**:

- High PL_ratio
- Low AVG_item_PL

Customers show **high dependence** on **private labels** but make fewer/smaller purchases.

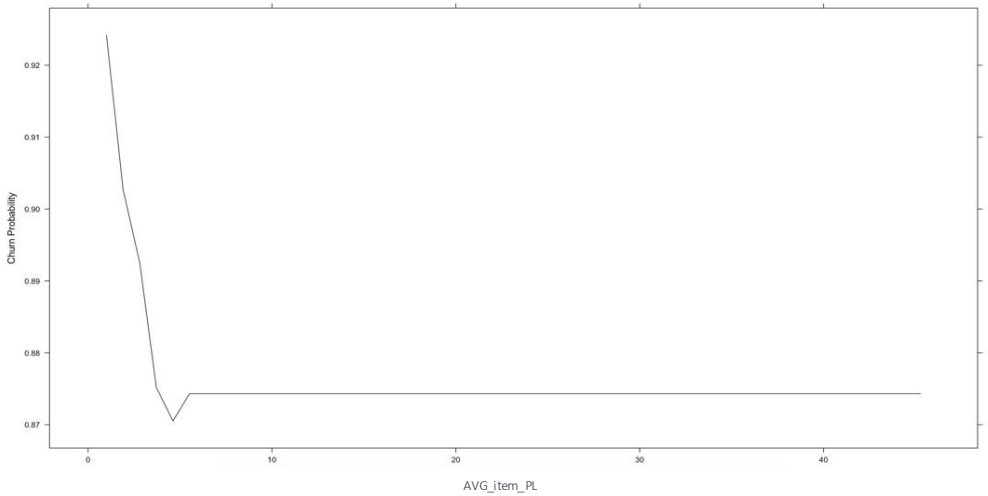
Risk: High churn probability due to **limited engagement** beyond private labels.

- Moderate PL_ratio
- High AVG_item_PL

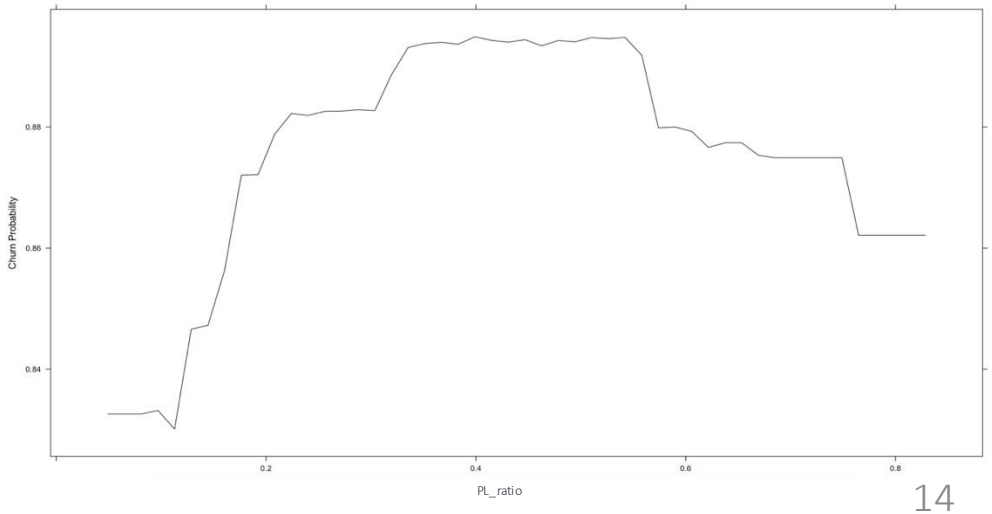
Customers purchase private label products in higher volumes, increasing also the overall purchase basket.

Impact: Lower churn probability, indicating stronger loyalty and engagement.

PDP for **AVG_item_PL**



PDP for **PL_ratio**



MARKETING STRATEGIES | 1

From the performed MBA analysis and the obtained results, we suggest following **activities to be implemented**, which would help to improve the consumer-related metrics and possibly attract new customers:

1. Creation of breakfast/lunch/dinner/aperitif baskets that provide ingredients for a full meal

WHY

Many rules in the obtained results are a **pair of products that are aimed to be used together** for creation of a certain common meal.

⇒ Providing customers with ready baskets created based on these rules can foster the purchase of certain products.

HOW

For each meal type (breakfast/lunch/dinner/aperitif) there are going to be **different options of products combinations** that combined give a certain dish; for each component of the dish there will be **different alternatives to choose from** - the same product from different producers/brands or different types of products that are used in similar ways; customers will be informed about this offer both in **physical store and on e-com platform** and they will be incentivized to take advantage of it by **the possibility to gain additional loyalty points**.

WHAT

The following **product combinations** are to be included:

- Biscuits – Milk
- Conserved tomatoes – Pasta
- Eggs – Vegetables
- Bananas/Cereals – Yogurt
- Ready sauces – Pasta
- Drinks – Salty snacks
- Frozen fish – Frozen vegetables
- Fresh Cheese – Cold Salami Cuts – Bread and alternatives
- Ready baking mixes – Flour – Milk
- Frozen bakery – Drinks

MARKETING STRATEGIES | 2

2. Creation of house cleaning baskets that provide full ready set of items for cleaning

3. Formation of standard personal care and hygiene baskets that contain set of basic everyday use products

WE ARE HERE

TO HELP CREATE YOUR BASKET

WHY

Several rules from the results show that items from these two categories **tend to be bought together with other items from the same category** → providing customers with ready baskets created based on these rules can foster the purchase of certain products;

HOW

Same approach as the one for meal baskets (explained in the previous pin-point) to be implemented;

WHAT

The following **product combinations** are to be included:

- House cleaning accessories – Dishes cleanser, Surfaces detergent
- Kitchen paper – House cleaning accessories, Toilet paper, Dishes detergent
- Fabric care – Fabric detergent
- Toiletries – Oral hygiene
- Body/Face treatment – Oral hygiene
- Haircare products – Toiletries
- Tablecloths – House cleaning accessories
- Surfaces detergent – Fabric care

&

!

* Additional possible strategy: **Bundled Discounts**

- Essence: offering loyalty points for **category-diverse shopping baskets**.
- Example: "Buy groceries, personal care, and household goods together to get 50 points for your next shopping!"

1. Reactivation and Retention Campaigns

Objective

Re-engage disengaged customers and gradually increase their shopping frequency and monetary contributions.

Strategy

1. Personalized Reactivation Offers
   - Target customers with high recency and low frequency using personalized discounts, using app.
   - Example: “We miss you! Earn double loyalty points on your next purchase this week.”
2. Gamified Incentives for Frequency
   - Encourage repeat shopping through challenges or milestones using the ecommerce and the proprietary application.
   - Example: “Make 5 purchases this month and earn 50 bonus loyalty points!”
3. Milestone-Based Rewards
   - Celebrate tenure milestones to create emotional connection and boost engagement, through the utilization of the app.
   - Example: “Happy 3 months with COOP! Enjoy 15% off your next purchase.”

Universal Application

- Low Engagement Customers: Use basic reactivation offers and low-threshold milestones.
- Consistently Moderate Customers: Add gamification to encourage frequent purchases.

KPIs

- Increase purchase frequency by 10-15%.
- Reactivate 30% of customers with high recency scores.

2. Strengthen Private Label Engagement

Objective

Use private labels as a loyalty driver, balancing volume and satisfaction to reduce churn.

Strategy

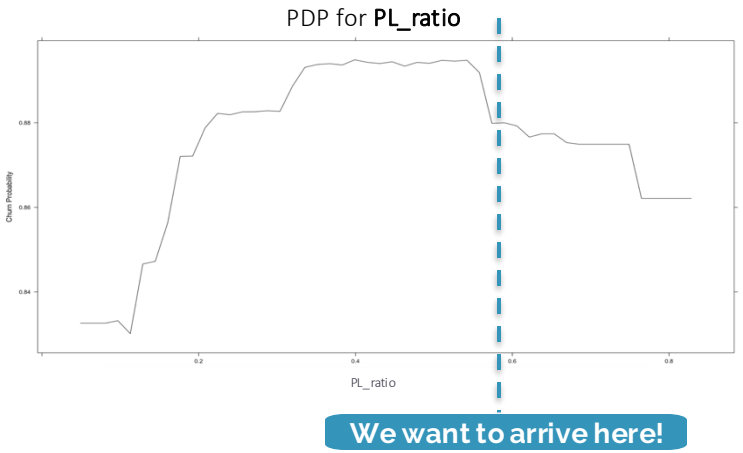
1. Volume-Based Discounts
   - Encourage larger purchases of PL item with discounts.
   - Example: “Buy 3 or more private label items and get 20% off.”
2. Gamified Private Label Challenges
   - Introduce a challenge to boost private label purchases.
   - Example: “Buy 10 private label items this purchase and earn 50 bonus points!”
3. Exclusive Private Label Rewards
   - Provide exclusive benefits for loyal private label customers.
   - Example: limited-edition products.

Universal Application

- Moderate PL_ratio Customers: Focus on increasing transaction volumes with targeted promotions.
- High PL_ratio Customers: Offer exclusive benefits like private label subscriptions or personalized perks to reinforce satisfaction.

KPIs

- Increase average `PL_ratio` by 5%.
- Boost `AVG_item_PL_per_transaction` by 10%.



3. Reward and Deepen Long-Term Loyalty

Objective

Retain long-tenure customers by rewarding commitment and deepening engagement.

Strategy

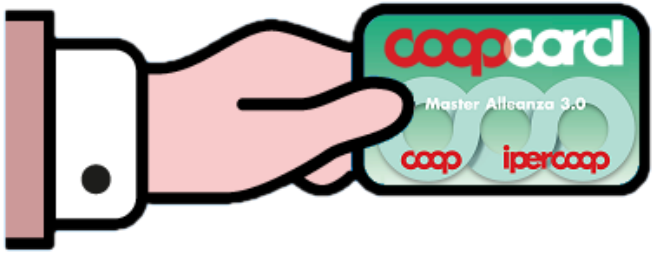
1. Exclusive Loyalty Tiers
   - Introduce premium tiers in the loyalty program based on tenure and spending.
   - Example: “Gold members receive free delivery for purchases over €50.”
2. Anniversary Rewards
   - Celebrate loyalty anniversaries with personalized offers.
   - Example: “Celebrate 1 year with COOP! Here’s 20% off your next order.”
3. Personalized Spending-Based Rewards
   - Tailor rewards to align with high-value spending patterns.
   - Example: “Spend €100 this month and earn 200 bonus points.”

Universal Application

- Super Customers: Offer exclusive benefits like VIP access and premium services.
- Moderately Engaged Customers: Provide incentives to move into higher engagement tiers.

KPIs

- Retain 95% of long-tenure customers (>12 months).
- Increase loyalty tier participation by 20%.



ANNEX | 1

Rules	LHS	RHS	Support	Confidence	Coverage	Lift	Count
{ACCESSORI GATTO} => {ALIMENTI PER GATTI}	{ACCESSORI GATTO}	{ALIMENTI PER GATTI}	0,02273369	0,70614035	0,0321943	7,91279425	483
{CAVOLI,CIPOLLE E AGLIO,MELE} => {ALTRE VERDURE}	{CAVOLI,CIPOLLE E AGLIO,MELE}	{ALTRE VERDURE}	0,02522828	0,69791667	0,03614798	4,66874606	536
{CAVOLI,RADICCHI} => {ALTRE VERDURE}	{CAVOLI,RADICCHI}	{ALTRE VERDURE}	0,02033324	0,68571429	0,02965264	4,58711767	432
{ALTRI ORTAGGI,GRASSI,RADICCHI} => {ALTRE VERDURE}	{ALTRI ORTAGGI,GRASSI,RADICCHI}	{ALTRE VERDURE}	0,02014497	0,67828843	0,02969971	4,53744207	428
{CAVOLI,FORMAGGI STAGIONATI SERVIZIO ASSISTITO,MELE} => {ALTRE VERDURE}	{CAVOLI,FORMAGGI STAGIONATI SERVIZIO ASSISTITO,MELE}	{ALTRE VERDURE}	0,02513414	0,67594937	0,03718347	4,52179479	534
{ALTRI ORTAGGI,CAVOLI,CIPOLLE E AGLIO} => {ALTRE VERDURE}	{ALTRI ORTAGGI,CAVOLI,CIPOLLE E AGLIO}	{ALTRE VERDURE}	0,03271204	0,67410281	0,04852678	4,50944218	695
{ALTRE VERDURE,CAVOLI,PATATE} => {CIPOLLE E AGLIO}	{ALTRE VERDURE,CAVOLI,PATATE}	{CIPOLLE E AGLIO}	0,02442813	0,65037594	0,03756001	4,16076098	519
{GRASSI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI} => {FARINA}	{GRASSI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI}	{FARINA}	0,02231008	0,69399707	0,03214723	3,49399568	474
{PANE E SOSTITUTIVI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI} => {FARINA}	{PANE E SOSTITUTIVI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI}	{FARINA}	0,02358091	0,69103448	0,03412407	3,47908024	501
{PASTA DI SEMOLA,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI} => {FARINA}	{PASTA DI SEMOLA,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI}	{FARINA}	0,021557	0,6887218	0,03130001	3,46743684	458
{INSAPORITORI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI} => {FARINA}	{INSAPORITORI,PREPARATI PER DOLCI,ZUCCHERO E DOLCIFICANTI}	{FARINA}	0,02000377	0,68770227	0,02908783	3,46230387	425
{GRASSI,PATATE,PREPARATI PER DOLCI} => {FARINA}	{GRASSI,PATATE,PREPARATI PER DOLCI}	{FARINA}	0,02122753	0,68645358	0,03092347	3,45601723	451
{CONDIMENTI PRONTI,IGIENE ORALE,PROD.PER CAPELLI} => {PRODOTTI BAGNO}	{CONDIMENTI PRONTI,IGIENE ORALE,PROD.PER CAPELLI}	{PRODOTTI BAGNO}	0,02038031	0,69614148	0,0292761	3,7557699	433
{COMPLETAMENTO IGIENE PERSONA,DETERGENZA TESSUTI,PROD.PER CAPELLI} => {PRODOTTI BAGNO}	{COMPLETAMENTO IGIENE PERSONA,DETERGENZA TESSUTI,PROD.PER CAPELLI}	{PRODOTTI BAGNO}	0,02146286	0,68778281	0,03120587	3,71067382	456
{AVVOLGENTI CUCINA,IGIENE ORALE,PROD.PER CAPELLI} => {PRODOTTI BAGNO}	{AVVOLGENTI CUCINA,IGIENE ORALE,PROD.PER CAPELLI}	{PRODOTTI BAGNO}	0,02118046	0,68597561	0,0308764	3,70092377	450
{AVVOLGENTI CUCINA,CARTA IGIENICA,PROD.PER CAPELLI} => {PRODOTTI BAGNO}	{AVVOLGENTI CUCINA,CARTA IGIENICA,PROD.PER CAPELLI}	{PRODOTTI BAGNO}	0,02033324	0,68031496	0,02988798	3,67038386	432
{CARTA IGIENICA,IGIENE ORALE,PROD.PER CAPELLI} => {PRODOTTI BAGNO}	{CARTA IGIENICA,IGIENE ORALE,PROD.PER CAPELLI}	{PRODOTTI BAGNO}	0,0323355	0,6761811	0,04782077	3,64808118	687
{BANANE,CIPOLLE E AGLIO,ZUCCHERO E DOLCIFICANTI} => {PATATE}	{BANANE,CIPOLLE E AGLIO,ZUCCHERO E DOLCIFICANTI}	{PATATE}	0,02245128	0,72824427	0,03082933	3,67949533	477
{BANANE,CIPOLLE E AGLIO,FARINA} => {PATATE}	{BANANE,CIPOLLE E AGLIO,FARINA}	{PATATE}	0,02391038	0,7267525	0,03290031	3,67195807	508
{ALTRE VERDURE,CIPOLLE E AGLIO,POLLO L.S.} => {PATATE}	{ALTRE VERDURE,CIPOLLE E AGLIO,POLLO L.S.}	{PATATE}	0,02541655	0,72	0,03530076	3,63784067	540

ANNEX | 2

lhs_item	associated_rhs	count_rhs
AGRUMI	ALTRI ORTAGGI, FORMAGGI STAGIONATI SERVIZIO ASSISTITO, LATTE, PASTA DI SEMOLA, PASTA FRESCA, SALUMI LIBERO SERVIZIO, SURGELATI VEGETALI, UOVA	8
ACCESSORI PULIZIA CASA	BISCOTTI, DETERG. STOVIGLIE, DETERGENTI SUPERFICI, GRANA LIBERO SERVIZIO, PASTA DI SEMOLA, PASTICCERIA INDUSTRIALE	6
ASCIUGATUTTO	ACCESSORI PULIZIA CASA, ACQUE, CARTA IGIENICA, DETERG. STOVIGLIE, DETERGENZA TESSUTI	5
ALTRE VERDURE	CIPOLLE E AGLIO, CONSERVE POMODORO, MELE, PATATE	4
AVVOLGENTI CUCINA	ACCESSORI PULIZIA CASA, CARTA IGIENICA, IGIENE ORALE, PRODOTTI BAGNO	4
PATATE	ALTRI ORTAGGI, BANANE, CIPOLLE E AGLIO, GRASSI	4
ALTRI ORTAGGI	ALTRE VERDURE, BISCOTTI, POLLO L.S.	3
BANANE	GRASSI, PATATE, YOGURT	3
BISCOTTI	AGRUMI, FORMAGGI STAGIONATI SERVIZIO ASSISTITO, SALUMI AFFETTATI TAKE AWAY	3
INSAPORITORI	BANANE, CONSERVE POMODORO, FORMAGGI FRESCHI	3
AROMI FRESCHI E PIANTINE DA CONSUMO	AGRUMI, ALTRI ORTAGGI	2
BIBITE	ACQUE, SNACK SALATI	2
BIRRA	SNACK SALATI	2
CARTA IGIENICA	CONSERVE DI PESCE, INSAPORITORI	2
CAVOLI	ALTRE VERDURE, CIPOLLE E AGLIO	2
CEREALI PRIMA COLAZIONE	YOGURT	2
CONDIMENTI	PASTA FRESCA	2
CONDIMENTI PRONTI	GRANA LIBERO SERVIZIO, PANE E SOSTITUTIVI	2
CURA TESSUTI	DETERGENZA TESSUTI	2
DESSERT	MERENDINE, PRODOTTI BASE CIOCCOLATO	2
DETERGENZA TESSUTI	CURA TESSUTI, DETERGENTI SUPERFICI	2
FORMAGGI STAGIONATI	GASTRONOMIA BASE FARINA, GRANA LIBERO SERVIZIO	2
PERE	MELE	2
POLLO L.S.	CONSERVE DI PESCE, UOVA	2

lhs_item	associated_rhs	count_rhs
POMODORI	VERDURE PREPARATE PRODUZIONE ESTERNA	2
PREPARATI PER DOLCI	FARINA, LATTE	2
PRODOTTI BAGNO	ACQUE, IGIENE ORALE	2
PRODOTTI BASE CIOCCOLATO	FORMAGGI FRESCHI, PASTICCERIA INDUSTRIALE	2
SALUMI A SERVIZIO ASSISTITO	PANE PROD. INTERNA	2
SOTTOLI / SOTTACETI(CONTORNI)	CONSERVE DI PESCE, FORMAGGI STAGIONATI SERVIZIO ASSISTITO	2
SPECIALITA' BASE PANE PROD. INTERNA	PANE PROD. INTERNA, VERDURE PREPARATE PRODUZIONE ESTERNA	2
SUCCHI DI FRUTTA	BIBITE, MERENDINE	2
SURGELATI BASE FARINA	BIBITE, SURGELATI VEGETALI	2
TRATTAMENTO CORPO/VISO	DETERGENTI SUPERFICI, IGIENE ORALE	2
VITELLO L.S.	PASTA DI SEMOLA, POLLO L.S.	2
ZUCCHERO E DOLCIFICANTI	FARINA, INSAPORITORI	2
ACCESSORI GATTO	ALIMENTI PER GATTI	1
ACETI/ALTRI CONDIMENTI	LATTE	1
ALIMENTI PREPARATI	PANE E SOSTITUTIVI	1
CARAMELLE/PRODOTTI BASE ZUCCHERO	PRODOTTI BASE CIOCCOLATO	1
CIPOLLE E AGLIO	PATATE	1
COMPLETAMENTO IGIENE PERSONA	CARTA IGIENICA	1
CONSERVE DI PESCE	SALUMI LIBERO SERVIZIO	1
CONSERVE POMODORO	AGRUMI	1
CREME SPALMABILI	MERENDINE	1
CUCINA SERVIZIO ASSISTITO	SALUMI AFFETTATI TAKE AWAY	1
DETERGENTI SUPERFICI	CURA TESSUTI	1

lhs_item	associated_rhs	count_rhs
FORMAGGI FRESCHI	UOVA	1
FORMAGGI STAGIONATI SERVIZIO ASSISTITO	SALUMI AFFETTATI TAKE AWAY	1
GASTRONOMIA PESCE	GASTRONOMIA BASE FARINA	1
GRANA A SERVIZIO ASSISTITO	FORMAGGI FRESCHI	1
GRASSI	FARINA	1
IGIENE ORALE	PRODOTTI BAGNO	1
INTIMI/ASSORBENTI	CURA TESSUTI	1
KIWI	BANANE	1
LATTE	PANE E SOSTITUTIVI	1
MELE	ALTRE VERDURE	1
MERENDINE	PASTICCERIA INDUSTRIALE	1
OLIO DI SEMI	GRASSI	1
PANE E SOSTITUTIVI	POLLO L.S.	1
PASTA ALL'UOVO	BISCOTTI	1
PASTA DI SEMOLA	PRODOTTI BASE CIOCCOLATO	1
PIATTI PRONTI SURGELATI	SURGELATI VEGETALI	1
PROD.PER CAPELLI	PRODOTTI BAGNO	1
RISO	DETERG. STOVIGLIE	1
SNACK SALATI	BIBITE	1
SURGELATI PESCE	GASTRONOMIA BASE FARINA	1
TOVAGLIE E TOVAGLIOLI	ACCESSORI PULIZIA CASA	1
VINI DA TAVOLA	INSAPORITORI	1
VITELLONE L.S.	CONSERVE POMODORO	1
WURSTEL	SALUMI LIBERO SERVIZIO	1